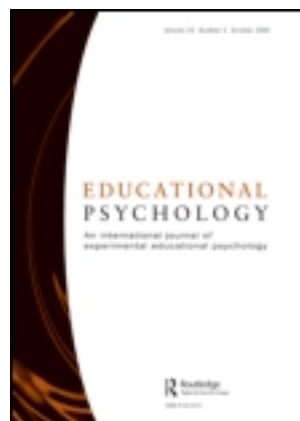


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On the Nature of Cognitive Style

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ABSTRACT *This paper reviews the origins of the construct of cognitive style, its fundamental dimensions and their method of assessment. The evidence for the independence of the style dimensions from one another, from intelligence and from personality is presented. The relationship of style to observed behaviours, such as learning performance, learning preferences, subject preferences and social behaviour, is described. Physiological measures and their relationship to style are considered. The distinction between style, which is likely to be relatively fixed, and strategies which are capable of being learned and developed, is discussed. A 'level' model is then outlined with, at the primary level: experience, personality sources and gender. Styles operate at the next level of cognitive control. The outer output level comprises the learning strategies. Priorities for further research are outlined.*

This paper will consider the nature of the construct of cognitive style, its assessment and its relationship to behaviour.

Cognitive Style and its Assessment

The Origins of the Construct of Cognitive Style

Early in the scientific study of psychology, attention was given to the notion that some people have a predominantly verbal way of representing information in thought, while others are more visual or imaginal (e.g. Galton, 1883; James, 1890) and this was continued by Bartlett (1932).

Between the early 1940s and the 1980s, various investigators observed what they felt represented style dimensions. However, the researchers generally worked in their own contexts, in isolation from one another, developed their own instruments for assessment and gave their own labels to the style they were studying with little reference to the work of others. Not surprisingly, this led to the development of a large variety of style labels. A number of workers suggested that many of these were but different conceptions of the same dimensions (e.g. Coan, 1974; Fowler, 1980; Brumby, 1982;

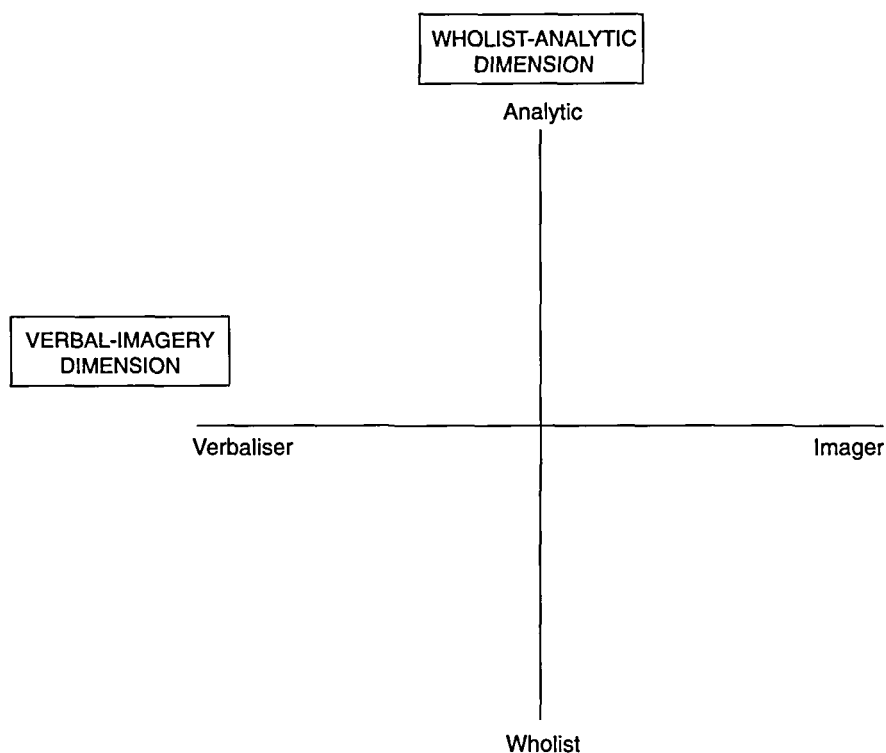


FIG. 1.

Miller, 1987; Riding & Buckle, 1990). Riding and Cheema (1991) found over 30 labels and, after reviewing the descriptions, correlations, methods of assessment and effect on behaviour, concluded that they could be grouped into two principal cognitive style dimension: the Wholist-Analytic and the Verbal-Imagery. A further review by Rayner and Riding (this issue) supported this conclusion.

The two basic dimensions of cognitive style may be summarised as follows:

- (1) The *Wholist-Analytic Style* dimension of whether an individual tends to *process* information in wholes or parts.
- (2) The *Verbal-Imagery Style* dimension of whether an individual is inclined to *represent* information during thinking verbally or in mental pictures.

Cognitive style has been defined by Tennant (1988) "an individual's characteristic and consistent approach to organising and processing information".

Style Versus Ability

An important aspect of this notion is that of style in contrast to simple ability. What are the characteristics of style that distinguish it from ability? Both style and ability will affect performance on a given task. The basic distinction between them is that performance on all tasks will improve as ability increases, whereas the effect of style on

performance for an individual will either be positive or negative depending on the nature of the task. It follows from this that for an individual at one end of a style dimension, a task of a type they find difficult will be found easier by someone at the other end of the dimension, and vice versa. For instance, if the dimension were the Verbal-Imagery style, then Verbalisers would find pictorial tasks more difficult than would Imagers, but they would find highly verbal tasks easier than would Imagers. In other words, in terms of style a person is *both* good *and* poor at tasks depending on the nature of the task, while for intelligence, they are *either* good *or* poor.

Assessment of Cognitive Style

Two general approaches to assessment have been used to study style: (a) introspective self-report and (b) tests of information processing, on the assumption that performance is facilitated by style.

(a) *Self-report.* A questionnaire, the Cognitive Style Index (Allinson & Hayes, 1996), probably assesses a dimension related to the Wholist-Analytic dimension. For the Verbal-Imagery dimension, Paivio (1971) devised the Individual Differences Questionnaire comprising 86 true/false item statements (47 for assessing the strength of preference in the verbal mode and 39 for the imagery mode). A development of this using differences in lateral eye movement was the Verbalizer-Visualizer Questionnaire (see, for instance, Richardson, 1994, pp. 35–37).

However, introspective self-report measures have inherent weaknesses. These include the subject's (i) possible inability to report their behaviour accurately and objectively, (ii) unwillingness to make the necessary effort to respond accurately and (iii) bias due to the pressure of social desirability in making responses (see, for instance, Kline, 1995, p. 512).

(b) *Measurement of Information Processing.* In terms of the Wholist-Analytic dimension, Witkin *et al.* (1971) produced the Embedded Figures Test (EFT), which required subjects to find simple geometrical figures within more complex ones in a given time. This was designed to assess field-dependence-independence and it was assumed that field-independent individuals would be able to do items more quickly than field-dependent ones. However, no subtest was used that the field-dependent individuals were likely to perform better on than the field-independent ones. As overall performance will also be affected by general ability or 'intelligence', it may be objected that the EFT assesses intelligence rather than style (see, for instance, Goldstein & Blackman, 1978, pp. 183–184; Flexer & Roberge, 1980; Riding & Pearson, 1994).

With the Verbal-Imagery dimension, Delaney (1978) used tests of both verbal and spatial ability. However, the results were used separately and not in combination, by means of a ratio or a difference.

The approach used by Riding (1991a) in developing the Cognitive Styles Analysis (CSA) was to assess performance on simple tasks that might then be representative of processing generally, with the intention of measuring an individual's position on both the Wholist-Analytic and the Verbal-Imagery dimensions. This was a development of the earlier work of Riding and Taylor (1976), Riding and Ashmore (1980), Riding and Calvey (1981) and Riding *et al.* (1989). The background to the development of the Cognitive Styles Analysis is given in Riding and Cheema (1991).

The CSA is computer presented and directly assesses both ends of the Wholist-Analytic and Verbal-Imagery dimensions, and comprises three subtests. The first assesses the Verbal-Imagery dimension by presenting 48 statements one at a time to be judged true or false. Half of the statements contain information about conceptual categories, while the rest describe the appearance of items. Half of the statements of each type are true. It was assumed that Imagers would respond more quickly to the appearance statements, because the objects could be readily represented as mental pictures and the information for the comparison could be obtained directly and rapidly from these images. In the case of the conceptual category items, it was assumed that Verbalisers would have a shorter response time because the semantic conceptual category membership is verbally abstract in nature and cannot be represented in visual form. The computer records the response time to each statement and calculates the Verbal-Imagery Ratio. A low ratio indicates a Verbaliser and a high ratio an Imager, with the intermediate position being described as Bimodal. It may be noted that in this approach individuals have to read both the verbal and the imagery items so that reading ability and reading speed are controlled for.

The second two subtests assess the Wholist-Analytic dimension. The first of these presents items containing pairs of complex geometrical figures which the individual is required to judge either the same or different. As this task involves judgements about the overall similarity of the two figures, it was assumed that a relatively fast response to this task would be possible by Wholists. The second presents items each comprising a simple geometrical shape (e.g. a square or a triangle) and a complex geometrical figure, and the individual is asked to indicate whether or not the simple shape is contained in the complex one by pressing one of the two marked response keys. This task requires a degree of disembedding of the simple shape from within the complex geometrical figure in order to establish that it is the same as the stimulus simple shape displayed: it was assumed that Analytics would be relatively quicker at this. Again, the computer records the latency of the responses and calculates the Wholist-Analytic Ratio. A low ratio corresponds to a Wholist and a high ratio to an Analytic. Ratios between these positions are labelled Intermediate. It should be noted that each style dimension is a continuum and subsequently labels will only be attached to ranges on a dimension for convenience.

The CSA provides a simple, quick and convenient means of assessing an individual's position on the two fundamental cognitive style dimensions (see Riding, 1991b). The instrument has several advantages:

- (a) It is an objective test, as defined by Cattell and Warburton (1967), as it is objectively scored and its method of assessment is not obviously apparent to those being assessed. It is consequently difficult for assesseees to contrive their results.
- (b) It positively assesses both ends of the style dimension. This is important, as otherwise it could be objected that the assessment is simply of ability.
- (c) As it does not contain questionnaire-type items, or difficult language, it can be used with a wide age-range from children to adults.
- (d) It is context free and can be used in a wide range of situations, such as schools, industry and the health service.
- (c) It is probably culture free in nature and it has been used in a number of countries [1].

Evidence for the Validity of Style as a Construct

For a style measure to be shown to possess a degree of construct validity, several conditions need to be fulfilled. These will include the following. The style dimensions should be:

- (1) independent of one another;
- (2) separate from intelligence;
- (3) independent of personality;
- (4) related to observed behaviours, such as (a) learning performance, (b) learning preferences, (c) subject preferences and (d) social behaviour;
- (5) related to physiological measures; and
- (6) such that they fulfil the requirements of a 'style'.

This consideration of style will be limited to studies using the Cognitive Styles Analysis approach [2].

(1) *The Independence of the Style Dimensions*

The two cognitive style dimensions are independent of one another, in that the position of an individual on one dimension does not affect their position on the other. Correlation between the two dimensions has been found to be consistently low and typically $r = \pm 0.1$ (e.g. Riding & Douglas 1993; Riding *et al.*, 1995).

Gender Differences. There do not appear to be overall gender differences with respect to cognitive style. Differences are usually small and nonsignificant on both dimensions ($p < 0.05$) (e.g. Riding *et al.*, 1995).

Age Differences. No longitudinal studies of the effect of age by assessing subjects at different ages have been undertaken. However, where samples were from a wide age-range, no significant correlation between age and style was observed. For example, with a sample of nurses aged 21–61 years, the correlations between age and Wholist-Analytic style and Verbal-Imager style, were, respectively, $r = 0.00$ and $r = 0.01$ (from the data collected by Riding & Wheeler, 1995).

(2) *Style and Intelligence*

Cognitive style appears to be independent of intelligence. Riding and Pearson (1994), with a sample of 119 12–13-year-old pupils, found that intelligence as measured by the subtests of the British Abilities Scale was not related to cognitive style. The correlations between the four subtests used and the Wholist-Analytic and Verbal-Imagery ratios were all low and nonsignificant and were, respectively: Digits – 0.01, 0.12; Similarities – 0.03, 0.01; Matrices – 0.10, 0.04; Speed of Information Processing 0.07, 0.02. Similar findings came from a Canadian study (Riding & Agrell, submitted) of the relationship between style and the Canadian Test of Cognitive Skills (CTCS) with 205 14–16-year-old students. The CTCS comprises four subtests containing intelligence test-type items. These together with the respective correlations with the Wholist-Analytic and Verbal-Imagery ratios were: Sequences 0.01, – 0.10; Analogies – 0.01, – 0.05; Memory – 0.03, – 0.03; Verbal Reasoning – 0.02 – 0.10. Again, all the correlations were very low. This study did, however, indicate an interaction between

style and intelligence in their effect on school achievement, such that style was more critical where pupils were of lower ability and the subject matter did not ideally suit their style. For instance, the difference in performance between high and low ability pupils was greater for the Analytic-Imagers than the Analytic-Verbalisers, where in the latter case that style was more naturally suited to learning academic subjects.

(3) *Personality and Style*

Riding and Wigley (submitted) in a study of 340 College of Further Education students aged 17–18 years used a range of questionnaire measures of personality and attitude, for comparison with cognitive style, and these included: (a) EPQ-R Short Scale (Eysenck & Eysenck, 1991) Extraversion, Neuroticism and Psychoticism; (b) IVE Questionnaire (Eysenck and Eysenck, 1991) Impulsiveness, Venturesomeness and Empathy, and (c) State and Trait Anxiety Inventory (Spielberger, 1977). A principal components factor analysis with oblique rotation gave four factors which were labelled: Anxiety, Impulsiveness, Empathy, and Style. No personality measure loaded beyond ± 0.33 on Style.

They also considered the interactive effect of Wholist-Analytic style and Verbal-Imagery style on personality measures. They found significant effects such that with Neuroticism, Wholist-Verbalisers and Analytic-Imagers were more anxious than Analytic-Verbalisers and Wholist-Imagers. With Impulsiveness, Wholist-Verbalisers and Analytic-Imagers were more impulsive/decisive than Analytic-Verbalisers and Wholist-Imagers. They proposed a tentative model in which physiologically based personality sources are independent of cognitive style but are moderated by style in their effect on behaviour.

(4) *Cognitive Styles and Behaviour*

The relationship between cognitive style and the following behaviours will be reviewed: (a) Learning Performance, (b) Learning Preferences, (c) Subject Preferences and (d) Social Behaviour.

(a) Style and Learning Performance. This section will consider the effect on learning performance of the interaction between style and the way the instructional material is structured, its mode of presentation, its type of content, as well as academic achievement.

The Structure of the Material. Several studies have found that an individual's position on the Wholist-Analytic dimension interacts with the structure of the learning material to affect performance. When presented with a prose passage for recall, Douglas and Riding (1993) found that with 77 11-year-old pupils, Wholists did best when the title of the passage was given before the passage was presented, rather than at the end, although this had little effect for Analytics. The result was interpreted as due to the Wholists being less able to structure material, being helped by a title at the beginning to give some organisation to the material.

Riding and Sadler-Smith (1992) compared performance on two versions of computer-presented instructional material on central heating systems with 129 14–19-year-old students. The two versions presented that same information, but one also contained an introduction to the five topics and each topic had a short overview at the start and

a summary at the end. In addition, there was an overall summary at the end of the five topics. This treatment slightly improved recall performance for Analytic-Verbalisers and Wholist-Imagers, but the additional material reduced performance for the other two styles, Wholist-Verbalisers and Analytic-Imagers. This was explained in terms of a lack on the part of the Analytic-Verbalisers of a facility to obtain a whole view, and of the Wholist-Imagers to be able readily to analyse the information into its structure. For these styles, the overview and summaries provided these to some extent. By contrast, the Wholist-Verbalisers and the Analytic-Imagers had some facility to do this for themselves and the additional material was therefore redundant and depressed performance.

Riding and Al-Sanabani (in prep.) with a sample of 200 10-15-year-old pupils studied the effect on recall performance of a short summary placed either before or after a main prose passage. The presence of a summary helped both Verbalisers and Imagers. However, for the younger pupils (10-12 years), the Verbalisers were most helped with the summary before, and the Imagers with it after. By contrast, with the older pupils (13-15 years) the Imagers gained most by having the summary first, and the Verbalisers with it second. Put another way, there was only a relatively small improvement in performance with age for the Verbalisers on the material with a summary, while there was a much greater change with the Imagers. This shift may have been due to the ways in which the styles developed learning strategies with age and, particularly, that the Imagers had found ways of coping more effectively with verbal material.

Riding and Adams (in prep.) with 84 12-year-old pupils considered the effect of context generated by Direct and Indirect antecedents on the response time to judging statements. An example of both is shown below. For each condition, the sentences were computer presented one at a time. When the pupil had understood a sentence a key was pressed. In the case of the last sentence, the pupil indicated whether it was true or false by pressing one of two marked keys.

DIRECT ANTECEDENT

(1) June was happy lying
in the hot sun.

(2) As a result she got mildly
sunburnt

(3) The sun can burn people

INDIRECT ANTECEDENT

(1) June joined the other
bathers on the beach

They found that in the case of the Indirect and Direct antecedents, the Verbalisers were faster than the Imagers. The reason for this was probably that Imagers take longer than Verbalisers because their habitual mode of processing is to generate mental pictures and these do not help in judging the final statement: while for the Verbalisers, the semantic area related to the statement is likely to be activated, facilitating a quick response. In all cases, except that of the Analytic-Verbalisers in the Direct condition, the Analytics took longer than the Wholists, perhaps because their processing is more exhaustive and considers a wide range of options. However, for the Analytic-Verbalisers in the Direct condition, they would have been able to use the context to guide them quickly to the properties of the objects mentioned and they would have been able more quickly to make the logical inferences, as they were more focused. In a sense, they could more quickly disembed the main idea suggested by the context. By contrast, in the more

diffuse Indirect contexts the Analytics could have generated too many possibilities and, hence, were slower than the Wholists who probably just waited for the question.

In summary, Wholists will benefit from help in structuring the material, while Analytics should be able to impose their own structure upon it.

Mode of Presentation. Two modes of presentation are available: the verbal and the pictorial. Riding and Ashmore (1980) with 74 11-year-old pupils presented groups of Verbalisers and Imagers (as assessed by the earlier Verbal-Imagery Code Test) with either a verbal or a pictorial version of the same information. They found that Verbalisers were superior with the verbal version and Imagers when learning in the pictorial mode. Within the instructional context, while purely verbal presentation is often possible, an alternative purely pictorial version is rarely an option as some words will also be necessary. However, it will usually be feasible to present information in both modes to some extent. Riding and Douglas (1993), with 59 15–16-year-old students, found that the computer presentation of material on motor car braking systems in a text-plus-picture format facilitated the learning by Imagers compared to the same content in a text-plus-text version. They further found that at recall in the text-plus-picture condition 50% of the Imagers used illustrations as part of their answers compared to only 12% of the Verbalisers.

Generally, Imagers learn best from pictorial presentations, while Verbalisers learn best from verbal presentations. It is also likely that lecturers and teachers will reflect their own style in the ways in which the present information, such that, for instance, Verbalisers will use a highly verbal content, while Imagers will use pictures and diagrams to illustrate their words. This is worthy of further study.

Style and Representational Control. Riding and Dyer (1980) distinguished between two levels of control of verbal and imagery performance: voluntary and involuntary. They suggested that, in the former, the individual makes a deliberate and conscious decision to generate an image or verbal response, while, in the latter, the image or verbal representation occurs spontaneously without conscious effort. In the present context, it may be argued that Verbalisers do not use images greatly during involuntary information processing, although they can generate them successfully by conscious effort. By contrast, Imagers habitually use involuntary imagery as a means of representing information. Further, for Imagers, mental pictures are likely to be less stable than those produced by Verbalisers as they will be liable to interference and displacement by further involuntary intrusive images. It may follow that the converse applies to Verbalisers and Imagers with respect to verbal associations, and that Verbalisers find that verbal associations tend to be less stable for them than they would be when consciously evoked by Imagers.

Douglas and Riding (1994), with 71 11-year-old pupils, studied drawing performance. The pupils were asked to draw two pictures, one from memory and one from a picture. Judges rated the drawings of the Verbalisers as significantly better than those of the Imagers. It may be that the fluid mental pictures of the Imagers interfered with the external representations.

It may well follow from this that, in the case of individuals of the same ability and similar on the Wholist-Analytic dimension, the Verbalisers are superior in speech which benefits from involuntary verbal fluency and relatively inferior in writing where the lack of verbal control due to spontaneous involuntary verbal associations may be a hin-

drance. By contrast, Imagers may be superior in written mode, but inferior in the spoken mode. This possibility awaits investigation.

Type of Content. In the case of the type of content of learning material, studies of 11- and 12-year-old pupils indicated that Imagers recall highly visually descriptive text better than acoustically complex and unfamiliar text, while the reverse holds for Verbalisers (Riding & Dyer, 1980; Riding & Calvey, 1981). Initial reading performance, which is obviously a very verbal task, has been found to be superior in Verbalisers. Riding and Anstey (1982), with 40 7-year-old children, assessed reading accuracy and comprehension and found that both declined from Verbaliser to Imager. Riding and Mathias (1991), with 80 11-year-olds, observed that for reading accuracy this effect was still very pronounced for Wholists, where Wholist-Verbalisers showed much greater proficiency at reading compared to Wholist-Imagers.

In terms of content type, individuals appear to learn best when information can be readily translated into their preferred Verbal-Imagery mode of representation.

Educational Achievement. The study of educational achievement and style poses problems. These derive from the nature of the subject, the ways in which it has been taught and the methods of assessment. A subject usually varies considerably within itself in terms of the type of content and the range of processing required. The method of instruction will vary from teacher to teacher and probably be a reflection of their styles. Further, the form of assessment can be varying proportions of course work and unseen examination, and can also vary in mode (verbal/pictorial).

Riding and Caine (1993) looked at the performance of 182 pupils in the General Certificate of Secondary Education (GCSE) (the British public examination at 16 years) in Mathematics, English Language and French. Overall, the Intermediate-Bimodals did well in all three subjects. There were also significant interactions between style and subject on performance, but the patterns obtained were not easy to interpret, probably for the reasons given above.

(b) Style and Learning Preference. Two methods of assessing learning preferences have been used: (i) self-report and (ii) observed behaviour choices.

Self-report Measures. In an investigation by Riding and Read (1996), 78 12-year-old pupils were individually questioned about their preferences in the subjects of English language and science with respect to mode of working and social context. With mode of working, in the case of the higher ability pupils, Imagers, particularly if they are Wholists, reported that they used less writing and more pictures than Verbalisers, especially where the subject allowed, as in science. The tendency by Imagers to use pictures, and Verbalisers writing, increased with ability. There was evidence that lower ability pupils were more constrained by the usual format of the subject than were those of higher ability.

The preferences for group, pair and individual working were that all pupils reported that they preferred group or pair work to individual work, and group work was particularly liked by Wholists, and especially by lower ability Wholist-Imagers. Individual work was least disliked by Analytics, particularly in the case of higher ability Verbalisers.

Sadler-Smith and Riding (submitted), used a questionnaire approach to study instructional preferences in 245 university business studies students. In terms of locus

of control, the Analytics preferred to have control themselves rather than to be controlled, while the Wholists had no preference.

The self-report approach suffers from the limitations mentioned above in the section considering the assessment of style. While questionnaire approaches have some uses, where possible observation of real behaviour is likely to be more relevant.

Observed Behaviour Choices. In a study by Riding and Watts (this issue), 90 female 16-year-old pupils were told within their class groups that three versions of a sheet giving information on study skills had been prepared for them, and that each sheet contained the same information but that the formats were different. They were then invited to come one at a time to the front of the class and to take one of the versions from the teacher's desk. Their choices were noted by the teacher. The versions were unstructured-verbal (paragraphs, without headings), structured-verbal (paragraphs, each with a clear heading) and structured-pictorial (paragraphs, each with a clear heading and a pictorial icon depicting the activity placed in the left margin). No pupils chose the unstructured-verbal version. With the two structured versions, there was a significant interaction with Verbal-Imagery style with the majority of the Verbalisers selecting the verbal version and most of the Imagers the pictorial. On the Wholist-Analytic dimension, the pictorial version was more attractive to the Wholists, perhaps because it looked more 'lively', while with the Analytics there was a slight preference for the more 'neat and tidy' verbal format. The results suggested that pupils are attracted to, and prefer to select, materials that appear to suit their own style.

(c) Subject Preferences. Riding and Staley (submitted) gave a questionnaire to 204 university business studies students which assessed their preferences about the content and presentation of their courses. For analysis the courses were grouped as, Information Technology, Accounting (accountancy and quantitative methods, as both of these composed largely of numerical concepts and computation) and Management (organisational behaviour and business workshop, both of which were concerned with the running of organisations). On the Wholist-Analytic dimension, Intermediates were least happy with Accounting and Information Technology and most happy with Management. For Accountancy and Information Technology, in terms of being positive about the courses, having a clear overview of the courses, least needing content overviews, finding the presentation order logical and liking the tutor's style, Wholists were highest in their ratings, followed by the Analytics, with the Intermediates having the lowest ratings. Accounting and Information Technology requires, on the one hand, the capacity for an overall view of the processes and their purposes, and on the other, the methodological following of rules. Wholists fit the first requirement and Analytics the latter. Intermediates tend to *alternate* between both, which is useful for a more creative approach, but does not *consistently* give either.

With Management, it was Intermediates who were most positive, generally found the presentation order logical and most liked the tutor's style. Wholists, in particular, rated these less highly. Management requires a more creative approach, requiring more problem-solving activities, which suits the Intermediates more than the Wholists.

With respect to the Wholist-Analytic dimension, when performance was considered it was noted that individuals underestimated their performance on the subjects that did not suit their style and overestimated it on those that did.

(d) Style and Social Behaviour. As cognitive style affects the ways in which an individual thinks about and internally represents situations in the external world, it is reasonable to expect that it might also be related to aspects of social behaviour.

The Wholist-Analytic style was derived from a family of styles which includes field-dependence-independence. Witkin *et al.* (1979) considered the effects of this dimension on aspects of personality related to the perception of self and nonself. They argued that self-nonself segregation increases with field-independence (see also Korchin, 1986). Riding (1991b) has suggested that the Wholist-Analytic style will be reflected socially in such dimensions as being dependent-self reliant, flexible-consistent, realistic-idealistic, vague-organised. Wholists will tend to the first mentioned extreme and Analytics the other.

In line with Huckabee (1974) and Schwartz (1975), Riding and Wicks (1978) found results on learning tasks consistent with the interpretation that Verbalisers would be extraverts and Imagers introverts. With 12-year-old children, Riding and Dyer (1980) found a correlation of -0.67 for boys and -0.76 for girls between an earlier measure of Verbal-Imagery style (using the Verbal-Imagery Code Test) and introversion-extraversion. As discussed in the section on Personality and Style, it is, however, not simply that extraversion-introversion and Verbaliser-Imager are respectively the social and cognitive manifestations of the same underlying mechanism. There are, however, aspects of the Verbal-Imagery dimension that affect social behaviour, as this dimension influences the focus and type of an individual's activity, externally and stimulating in the case of Verbalisers, and internally and more passive for Imagers.

Social Relationships. Riding and Wright (1995) asked 149 undergraduate students living in university flats, with typically five per flat, to rate their flatmates in terms of personal characteristics. The Wholists were rated as more assertive, humorous and helpful, while the Analytics were seen as more shy. Further, the degree of unity in each flat was assessed by rating questions such as, "Relations in my accommodation have been harmonious." When compared to the majority style in each flat, the order of reported unity from least to highest was: Analytic-Imager, Wholist-Imager, Analytic-Verbaliser, Wholist-Verbaliser. This order was as expected as Analytics are likely to be more separate than Wholists and Imagers are likely to be more inward and detached than Verbalisers.

Riding *et al.* (1995) asked 12-year-old pupils to rate the children in their class in terms of being Active (outgoing, lively, humorous), Modest (shy, quiet) and Responsible (serious, patient). They found Verbalisers highest on Active, Bimodals highest on Modest and that Responsibility increased from Verbaliser to Imager.

Behaviour Problems. Riding and Craig (submitted) assessed the style of 83 10–18-year-old pupils in two residential special schools to which the pupils had been referred because of behaviour problems. The percentage of pupils in the Wholist-Analytic dimension style groupings as defined by a control sample was Wholists 46%, Intermediates 34% and Analytics 20%. Rayner and Riding (1996) studied 17 15–17-year-old pupils attending a Truancy Unit because of their previous failure to attend school. Again, comparison with a control sample showed a significant difference on the Wholist-Analytic dimension with Wholists 41%, Intermediates 53% and Analytics 6%. The high proportion of Wholists with problem behaviour might possibly be related to a lack of strong internal control on the part of Wholists if they are then in a situation

where there is insufficient structure and parental discipline within their home environment.

Stress and Aspect of Occupation. Borg and Riding (1993) gave a sample 212 Maltese secondary school teachers a stress questionnaire. This indicated that the amount of stress reported was in line with the expectation of Wholists being more open and people orientated, and Analytics being more structured and organised. The Wholists reported more stress from time pressures and staff relations, while the Analytics found pupil misbehaviour and working conditions more stressful.

Riding and Wheeler (1995), in a study of 204 nurses, found that Verbalisers had a significantly higher grade than Imagers. Further, they found evidence that the type of nursing that most suited nurses was related to their style. A questionnaire was used to assess their reported level of stress and of general satisfaction with their job. A Suitability Index was obtained from (satisfaction minus stress) and this was compared with their style and type of nursing. The styles with the highest and lowest indices on three types of nursing are shown below.

Suitability Level	Type of nursing and style		
	General	Orthopaedic	Psychiatric
Highest	Analytic-Verbaliser	Intermediate-Bimodals	Wholist-Imagers
Lowest	Wholist-Imager	Analytic-Imager	Intermediate-Verbaliser

It could be argued that General Nursing is dynamic, requiring conscientious activity, Orthopaedic nursing is slower needing patience, and Psychiatric Nursing is demanding benefiting from tolerance.

This approach could usefully be applied to other occupations.

(5) *Style and Physiological Mechanisms*

Riding *et al.* (1993) reviewed work on EEG and verbal and imagery processing. They suggested that for the Verbal-Imagery dimension, EEG alpha suppression (indicating mental activity) during information processing would occur over the left hemisphere for Verbalisers and over the right for Imagers. With respect to the Wholist-Analytic dimension, no clear prediction was made other than it would not be in terms of left-right hemispheric specialisation.

Riding *et al.* (1997) reported a study of EEG and style which recorded EEG alpha in 15 subjects ranging in age from 18 to 36 years, while they did computer-presented word targeting tasks in which words appeared one at a time on the screen at a rate of two, five and 10 words/second, and in pairs at five and 10 word-pairs/second. The task required subjects to monitor the displayed words and respond with a button press whenever a noun from the superordinate categories of 'fruit' or 'vegetable' was displayed (e.g. *apple* or *carrot*). This study showed a significant effect ($p = 0.002$) for the Wholist-Analytic dimension along the midline (locations Fz, Cz and Pz) with alpha suppression being greater for the Analytics and fairly uniform along the line, while for the Wholists suppression decreased from anterior to posterior. For the Verbal-Imagery

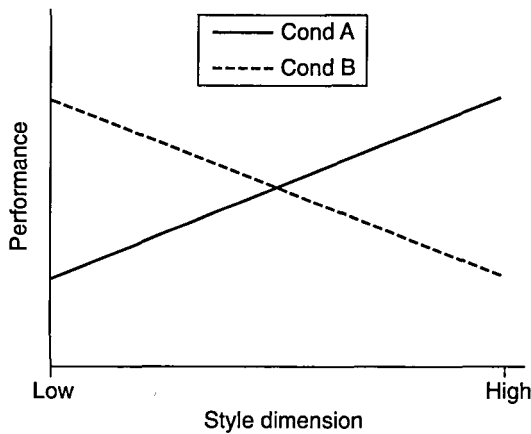


FIG. 2. Schematic graph of performance and condition.

dimension, the ratio of the power output at T5 (left) to that at T6 (right) correlated with the Verbal-Imagery ratio $r = 0.50$ ($p = 0.055$) in the expected direction. It should be noted that in this study the tasks were likely to favour Analytics and Verbalisers. In view of this, the finding of significant style effects is encouraging. Further studies are required using a range of information-processing tasks that are in the natural mode for each style.

(6) *Style Dimension Positive and Negative Interaction with a Variable*

It is in the nature of a style, as distinct from an ability, that it should interact with a variable such that the relative performance of an individual at one extreme of a dimension should be higher than that of a person at the other end in one condition, but that the situation should be reversed when the condition is changed. This is shown schematically in Fig. 2.

Fig. 2 shows the idealised relationship, which, in practice, is not likely to be so linear and the crossover may not be at the centre of the dimension. Further, in many real-life tasks there is likely to be an interaction between the two dimensions and the condition, rather than an effect of only one of the dimensions without any effect of the other. However, single dimension interactions have been observed. For the Wholist-Analytic dimension, this type of interaction was observed in the case of Douglas and Riding (1993) in the study of the effect of the position, before or after, of a prose passage title on recall. In the case of the Verbal-Imagery dimension, the interaction was found in the studies by Riding and Douglas (1993) on text-plus-text versus text-plus-picture presentation of learning material, and by Riding and Watts (this issue) in the observed preference for verbal or pictorial formats of instructional material.

Towards a Model of Style

Style and Strategies

It is useful to distinguish between *style* and *strategy*. Style probably has a physiological basis and is fairly fixed for the individual. By contrast, strategies are ways that may be learned and developed to cope with situations and tasks, and particularly methods of

utilising styles to make the best of situations for which they are not ideally suited. Within the literature, the term 'learning style' is sometimes used to refer to what here is considered to be a strategy. A useful distinction has been made by Curry (1983).

Curry's Onion Model

Curry (1983) proposed that all cognitive/learning style measures may be grouped into three main types or "strata resembling layers of an onion". By organising learning style measures like this, "learning behaviour is fundamentally controlled by the central personality dimension, translated through middle strata information processing dimensions and, given a final twist by interaction with environmental factors encountered in the outer strata".

According to Curry, the outermost layer of the onion and the most observable style is what she labels as "instructional preference". Examples of measures of this style are the Learning Preference Inventory (Rezler & Rezmovic, 1981) and the Grasha Reichmann Students Learning Styles Scales (Reichmann & Grasha, 1974). Instructional preference refers to the individual's choice of environment in which to learn. As this is the layer that is most exposed to the learner's environment, learner expectations, teacher expectations and other external features, Curry believes instructional preference to be the least stable, the most easily influenced level of measurement in the learning styles area.

The second layer of the onion model is referred to as the 'information processing style'. This is considered as the individual's intellectual approach to assimilating information, and, because this processing does not directly involve the environment, Curry believes that measures of this style are more stable than instructional preference, but are still modifiable by learning strategies. Measures such as the Learning Style Inventory (Kolb, 1976) and Inventory of Learning Processes (Schmeck *et al.*, 1977) are all thought to be dealing with information processing style concepts applicable at the intersection between fundamental personality level individual differences and environmentally offered learning format choices.

The third and the innermost layer of the hypothetical learning style onion is 'cognitive personality style'. The cognitive personality style is defined as the individual's approach to adapting and assimilating information, which does not interact directly with the environment, but is an underlying and relatively permanent personality dimension that is expressed indirectly and is apparent only when an individual's behaviour is observed across many learning instances. Examples of measures which Curry suggested assess this third type of learning style are the Embedded Figures Test (Witkin, 1962); Myers-Briggs Type Indicator (Myers, 1962); and Matching Familiar Figures Test (Kagan, 1965).

On the basis of the onion model, the Cognitive Styles Analysis would be placed in the innermost level.

The Cognitive Control Model

A possible model, which is an elaboration of that of Curry, of Riding and Rayner (1995) and Riding and Wigley (submitted), and which also bears some similarities to that of Furnham (1995, p. 398), will be sketched in outline and this is shown schematically in Fig. 3.

At the inmost level, there are a number of underlying primary sources comprising the

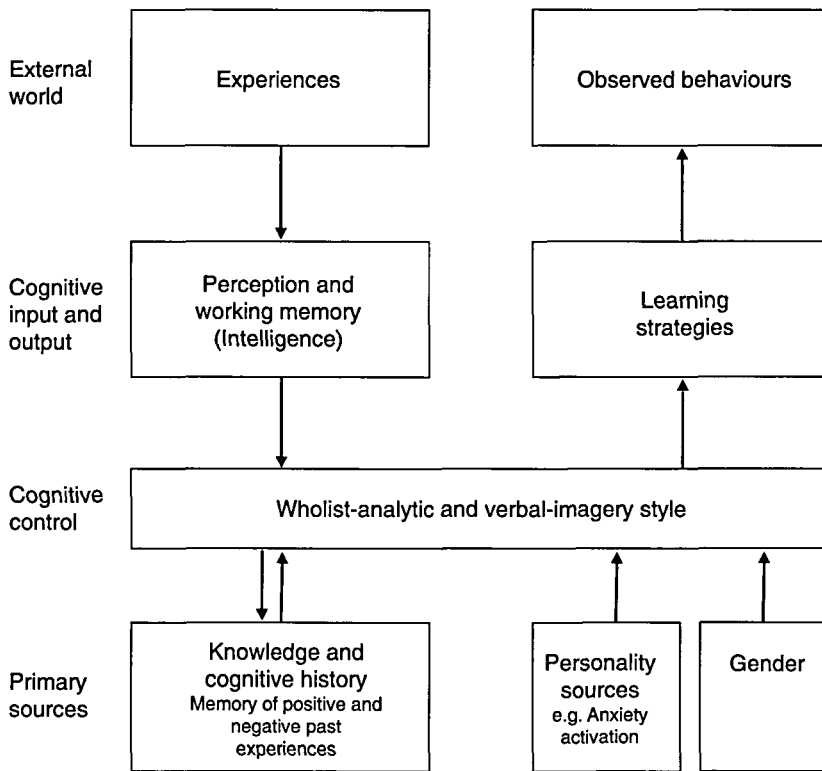


FIG. 3.

memory of the individual's past experiences and knowledge, their underlying personality sources and their gender. In order to make the model more concrete, some of the personality sources have been given tentative names, although clear physiological mechanisms for them have yet to be fully identified.

The next level is that of cognitive control. This comprises the two dimensions of style, the Wholist-Analytic and the Verbal-Imagery. It provides the organisational and representational interface between the internal state and the external world. It combines the internal state with the information from the external world and imposes on the response and view its own structure and form. The ways in which the Wholist-Analytic dimension and the Verbal-Imagery dimension may affect attitude and behaviour have been discussed by Riding (1991b) and Riding (1994).

At the input level will the perceptual and working memory processing system which analyses the incoming information. The performance of this may be considered to be the main determiner of 'intelligence' as assessed by intelligence tests, as many test items seem to concentrate on this area.

At the output level, there will be the learning strategies. These will have developed by, for instance, the individual then sensing that certain modes are easier to use, and then recognising a *learning preference*, and then deciding to 'translate' new incoming information into that representation. At this level, there will also be the cognitive response 'set' that influences social behaviour. (It may be observed that learning strategies are at a different level in this model from that of Curry.)

The perception of experiences will probably be moderated by the cognitive control

level in interaction with the cognitive history and the primary personality sources. Taking anxiety as an example, and building on Spielberger (1966) and Eysenck (1991), external stimuli are assessed by the cognitive control level as to whether or not they are threatening. Those that are interpreted as being threatening trigger the anxiety source and the level of state anxiety increases and this, in turn, is reflected in behaviour. The interpretation of the threat will be affected by cognitive style and also the characteristics of the trait anxiety source.

Some tentative evidence for the model comes from the observations of interactions between intelligence, style and gender in affecting academic achievement (Riding & Agrell (submitted)) and between style and personality sources in affecting social behaviour (Riding & Wigley (submitted)).

Conclusion

In order to pursue research, and particularly to develop a model and evaluate a construct, the researcher needs to have sufficient evidence to sustain belief that the construct may exist in order to maintain the energy to undertake the research. On the other hand, it is also necessary to have a degree of scepticism in order to retain the openness required to evaluate the findings. The psychological study of any construct poses difficulties. Even a generally accepted notion, such as intelligence, has been questioned as to whether it really exists as a valid construct (see, for instance, Howe 1990). Howe warns, "It is particularly important not to fall into the trap of believing that by describing someone's performance we are also providing the reasons for it" (1990, p. 493).

The validity of the construct of cognitive style, as assessed by the Cognitive Styles Analysis approach, is supported by evidence that the dimensions are independent of one another, separate from intelligence, independent of, but interacting with, personality; and related to observed behaviours, such as learning performance, learning preferences, subject preferences and social behaviour. There is also an indication of a relationship between style and physiological measures. In addition, cognitive style fulfils the requirement for the distinction between style and ability. Further, the observed relationships between the dimensions and behaviour are consistent with the view that the Wholist-Analytic dimension is concerned with organisation and the Verbal-Imagery dimension with representation.

Further Research

While there is evidence to support the notion of style, a number of questions remain and further research is needed to clarify these. Possible research areas include the following.

(a) *What Are the Origins of Style?* An obvious question concerns the origins of style. They are either inbuilt or develop with experience. It could be that an individual is born with a predisposition to use one mode of representation in preference to another. Are there style genes? On the other hand, there is the possibility that a style emerges from a difference between two complementary abilities. In order to illustrate this, one can deliberately oversimplify how the brain works and consider that within it there is one processor for verbal information and another for pictorial. If, in an individual, these two processors have unequal speed and processing capacity, then the individual will tend to

use the one in preference to the other whenever possible, and this will lead to the establishment of a style, or habitual preference for one over the other. Whatever the mechanism, style appears to be present at a relatively early age. Riding and Taylor (1976) found evidence that the Verbal-Imagery style was strongly apparent in 7-year-olds. Methods of assessing style in young children and infants need to be developed.

(b) Are There Other Fundamental Style Dimensions? Do the Wholist-Analytic and the Verbal-Imagery dimensions encompass all style descriptors or are there are possibly other dimensions? If there are, then they must fulfil the condition that for a style to exist as a dimension, the two processors must represent opposites in some sense. In the case of the Verbal-Imagery dimension, while words are not the opposite of pictures, they can be alternative ways of representing the same information. For the Wholist-Analytic dimension, it is possible to see a progression from the whole to the parts. The whole is, in this sense, the opposite of the parts; the wood comprises the individual trees. Studies are required to compare other existing measures of style and strategy, provided they have also been shown to possess a reasonable degree of construct validity, with the Wholist-Analytic and Verbal-Imagery dimensions.

(c) What Is the Relationship Between Physiological Measures and Style? The identification of the underlying physiological mechanisms associated with style is important. Further studies using more sophisticated brain-scanning techniques to compare neural activity during a range of tasks would be useful.

(d) Are There Cultural Differences in Style? Most of the studies using the Cognitive Styles Analysis approach have been in the UK. The possible effect of culture on style is of interest. If similar style patterns were to be demonstrated in a wide range of cultures, then this would contribute to the understanding of the nature of style. In English there are versions of the CSA for use in the UK, Northern America and Australia. In addition, French, Dutch, Malay and Arabic versions have been developed.

(e) What Is the Interaction Between Other Sources and Style? In considering the impact of styles on behaviour, such as learning performance, it is necessary to bear in mind the range of individual variables that might affect learning. For a particular task, these will include the individual's:

- (a) level of ability/intelligence;
- (b) present relevant knowledge which is necessary to give meaning to the new information, or, on the other hand, that could make the new learning redundant;
- (c) degree of motivation, including the perceived relevance of the task;
- (d) gender; and
- (e) style.

Clearly style will only be one of many influences. It will not show an effect if there is a lot of noise in the system, occurring because other relevant variables are not measured and included in the analysis. Further, style is not likely to be critical when the task is simple. It is likely to be important, however, where the learner is under pressure because, relative to their ability etc., the task is difficult. The investigation of the conditions in which style is most critical are required. Other research could usefully undertake the further study of personality descriptors and style.

With intelligence measures it may be noted that these are likely to include more than

just processing capacity and facility, as the items commonly used in intelligence tests also assess learned knowledge and the application of acquired skills. They therefore include within the measure the ability to learn and perform, and, hence, explain more of the variance than style, but they are not pure measures.

(f) *What Is the Interaction Between Other Experiences and Style?* The effect of previous experiences on style reactions is of interest. For example, does stress make Analytics more analytic and cautious and Wholists more impulsive?

(g) *What Are the Mechanisms for Strategy Development?* Given that a particular style finds certain tasks more difficult, how can the development of effective strategies be fostered in an individual? Various approach could be tried, for example, (i) pairing people of dissimilar style so that they learn from one another's approaches as they work together, (ii) deliberately present materials in mis-matched mode to encourage the production of methods of recoding, (iii) make people aware of the potential of the complementary nature of style dimensions where they exist in the case of Analytic-Imagers who could use their imaging facility to give an overall, 'whole' view which they lack as Analytics.

Further, the progression of strategy development could be studied by investigating the progression from awareness of ease of difficulty in doing a task, through preferences, awareness of limitations, to the development and refinement of strategies to cope with stylistically inappropriate modes or situations.

(h) *What is the Range of Practical Applications of Style?* Riding (1994) speculated on the practical work-place applications of style for personal development, team roles, management style and interpersonal relationships. Systematic investigation of the types of application would clarify the practical use of style in a range of work situations.

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NOTES

- [1] The Cognitive Styles Analysis is available in various languages: English versions for the UK, North American and Australasian contexts; French, Dutch, Arabic and Malay. Other language versions are in preparation.
- [2] In this review of work using the Cognitive Styles Analysis, the style dimension groupings (e.g. Wholist) were usually relative to the sample used and not absolute according to a fixed ratio range. Further, only two divisions were used where the sample was relatively small (e.g. Wholist and Analytic) and three where it was larger (e.g. Wholist, Intermediate and Analytic).

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